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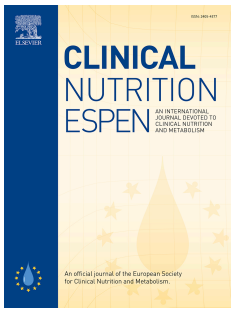

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PROSPERO registration number: CRD42024505377

Abstract:

Objective: This systematic review aimed to assess the effects of caffeine supplements on sleep parameters among professional athletes.

Methods: A systematic search of randomized controlled trials (PROSPERO: CRD42024505377) was performed from 1980 to December 2023 through Web of Science (ISI), Cinahl, Embase, CENTRAL, PubMed/MEDLINE, Scopus, and Scienceopen. The effect of caffeine supplement on all sleep parameters (e.g. duration, quality, insomnia), assessed through objective and subjective methods, was investigated among the athletic community.

Results: Of 1469 records, nine trials were eligible for the current review. The studies showed varying results concerning sleep quality, quantity, efficiency, number of awakenings, sleep onset latency, and other sleep-related variables. These differences in findings may be attributable to factors such as the timing of caffeine consumption in relation to sleep time and the time of exercise, habitual caffeine use, and the dose of caffeine prescribed. Given the nature of caffeine, insomnia following ingestion is likely to occur.

Conclusions: This review explores the mechanisms by which caffeine influences sleep in athletes. While caffeine supplementation may enhance athletic performance, it could have a detrimental effect on sleep and therefore recovery. It is important that supplementation considers individual responses to caffeine so that it does not adversely affect sleep in this population.

Keywords: Systematic Review; Sport; Caffeine; Coffee; Sleep; Insomnia

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Introduction

There has been an increased prevalence of caffeine consumption within the realm of sports, particularly since its removal from the prohibited substances list by the World Anti-Doping Agency (WADA) in 2004 (1, 2). Substantiated by empirical evidence, the utilisation of caffeine supplementation is prominently observed among athletes preceding or during competitive events (2-4). Although caffeine supplementation has become widespread across various sporting disciplines, there is a tendency for its use to be more common within endurance sports (2). However, with team sports scheduling now being subject to media timeframes, the use of caffeine is also increasing with later start times that coincide with prime-time audiences in team sports (5).

Further to this, there is some evidence that athletes may be consuming caffeinated supplements and caffeinated sports beverages more regularly, compared with non-athletic populations (1, 6). The apparent inclination of athletes to use caffeine may be ascribed, in large part, to the possibility of it benefiting sports and athletic performance (7, 8). This idea can be substantiated throughout a large body of scientific literature that underscores the positive ergogenic effects of caffeine (7-12).

Caffeine acts by antagonising adenosine receptors in the brain. Adenosine is a neuromodulator that promotes sleep and suppresses wakefulness. By blocking the adenosine receptors, caffeine is effective in reducing the feeling of tiredness and increasing alertness (13). These effects of caffeine supplementation have been demonstrated as being beneficial across diverse parameters associated with exercise performance, particularly in the domain of sports (7-12). Meta-analyses have highlighted the capacity of acute caffeine intake to enhance aerobic endurance (14, 15), anaerobic performance (9), maximal strength and muscular endurance (16, 17), resistance exercise movement velocity (18), as well as sprint and agility performance (19). The use and efficacy of caffeine in enhancing performance within team sports and combat sports has also been identified (8).

While there has been a plethora of research investigating the benefits of caffeine on athletic performance, there has been limited attention directed towards the adverse side effects of caffeine consumption, particularly in the context of athletes (20). It has been observed that there is a tendency for athletes to engage in moderate-to-high doses of caffeine intake during competitions (3, 5, 21) despite benefits to performance being attributed to lower doses (22, 23). Caffeine supplementation, even at low doses (≤3.0 mg/kg), can result in undesirable effects

such as sleeplessness, insomnia, and increased anxiety (20, 24). The frequency and severity of these negative effects tend to be contingent upon dosage levels. Higher doses of caffeine (~9.0 mg/kg) have been shown to result in a pronounced escalation of negative side-effects. These include increased anxiety, nervousness and agitation, gastrointestinal problems, particularly in individuals unaccustomed to caffeine intake and sleep quality including, sleep disturbances and insomnia (25).

Research has illustrated that sleep is vital for athletic performance (26, 27) and that sleep deficiency increases the risk of injuries in sport and reduces cognitive functioning associated with decision making in sport (26-28). Sleep has been identified as one of the most important factors in post-performance recovery (29-31). A lack of sleep has been shown to impair inflammatory and endocrine system responses in exercise-induced muscle damage (31), reduce submaximal performance in strength, whilst inadvertently increasing perceived effort (32) and impair an athlete's immune system (33). Considering the negative effects on sleep, the risk-benefit ratio of acute caffeine intake by athletes remains ambiguous.

It is contingent upon factors such as the nature of the sport, dosage administered, habitual daily caffeine intake, and individual variations in response to caffeine. High doses or unregulated supplementation before and during exercise may compromise rather than enhance performance, impeding post-exercise recovery. This is particularly pertinent in sports where athletes compete regularly within a shortened timeframe or have limited recovery time (29, 30).

The effects on sleep among athlete populations associated with the use of caffeine holds significance for formulating effective caffeine supplementation strategies. Supplementation strategies should both optimise athletic performance while safeguarding athletes' sleep and recovery. There remains an absence of a systematic and comprehensive study that summarises the literature on the effects of caffeine consumption in athletes and its effects on sleep. For this reason, this systematic review aims to comprehensively and critically assess the available randomised controlled trials (RCTs) that investigate the effects of caffeine on sleep among athlete populations and determine whether dietary supplementation with caffeine affects sleep in athletes.

Methods

Study Design

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This article is a systematic review focussing on caffeine supplementation and its effects on sleep in athletes. The study protocol was registered in the PROSPERO International prospective register of systematic reviews on January 2024 (CRD42024505377; available at https://www.crd.york.ac.uk/prospero/display_record.php?RecordID=505377). The systematic review of randomised controlled trials (RCTs) was conducted and is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (34). The main research question was: "Does dietary supplementation with caffeine affect sleep in athletes". The PICOS components of this systematic review is represented in **Table 1**.

Search strategy

The following databases were systematically searched from 1980 until the end of December 2023, with no language restriction.: Web of Science (ISI), Cinahl, PubMed/MEDLINE, Embase, Cochrane Central Registry of Controlled Trials (CENTRAL), Scopus, and Scienceopen. A hand search was performed to find the relevant studies from the reference list of the included studies, along with the related reviews, systematic reviews, and meta-analyses. A combination of MeSH terms used within the search included "caffeine" and "sleep". To reduce the search results, we included synonyms of "clinical trial" in the search syntax as well. The complete search strategy for all databases is presented in **Table S1**.

Screening and Study Selection

One investigator searched the databases, removed the duplicates, and screened the studies according to their title and abstract (ZSH). Two independent reviewers then performed the selection of the eligible studies (MB and CL) based on their full texts. The full text was sought for all relevant outcomes, and their eligibility was double checked according to the inclusion and exclusion criteria. Disagreement between the two investigators were resolved by consultation with a third (JB) reviewer.

Studies were included if they met the following inclusion criteria: (a) RCTs on; (b) examination of caffeine supplements in the form of powder, tablet or capsule and placebo; (c) adult athletes as participants >18 to <65 years of age; (d) reported at least one sleep-related outcome (objective methods: actigraphy or polysomnography; subjective methods: sleep quality, duration of sleep, sleep disturbances, insomnia). "Athlete" was defined as a person who competes in one or more sports involving physical strength, speed, power, or endurance. Studies were excluded if: (e) participants were not athletes; (f) participants were engaged in

recreational exercise and activities but not competing as an athlete as defined above; (g) the study design was unsuitable (non-randomised studies, observational studies, case reports, study protocols, editorials and reviews); (h) study full-text was not available (e.g. conference abstracts); (i) caffeine was administered in the form of energy drinks (which contained other nutrients and amino acids that might affect sleep); and (j) athlete participants were < 18 or > 65 years of age.

Data extraction:

Included studies were reviewed by two separate investigators (MB and CL). The data were extracted and tabulated including: first author name, publication year, country, total and per arm number of participants, gender, mean age, weight, study design, athletic population, prescribed physical activity, intervention in each arm, caffeine dose and frequency of intake, time of caffeine supplementation relative to physical activity, sleep parameters assessed and assessment tools.

Quality Assessment:

Two researchers categorised the primary studies included as low, high or uncertain risk of bias (MB and CL). Any disagreements were resolved by consulting a third researcher (ZSH). All of the included trials employed a cross-over design, therefore the Revised Cochrane risk-of-bias tool for randomized crossover trials (RoB-2) was used to assess the quality of eligible studies (35). The risk of bias tool covers six domains of bias. Risk of bias arising from: the randomization process (Domain 1), period and carryover effects (Domain S), deviations from the intended interventions (Domain 2), missing outcome data (Domain 3), measurement of the outcome (Domain 4), selection of the reported result (Domain 5). If the study was judged to be at low risk of bias for all domains, it was considered as “low risk”. If the study was judged to raise some concerns in at least one domain, but not to be at high risk of bias for any domain, it was considered as “some concerns”; and in case the study was judged to be at high risk of bias in at least one domain or had some concerns for multiple domains, it was categorized as “high risk” (35).

Risk of bias

As assessed by RoB-2, two studies were categorized as low risk, 2 unclear (36, 37), and 1 study as high risk of bias (38). Measurement of the outcome (domain 4) was the source of bias in two

studies (36, 38), and in two of the eligible studies, deviations from the intended intervention (domain 2) was determined as the source of bias (37, 38).

Data-analysis

To conduct a meta-analysis both pre- and post-intervention data were required. Some studies used objective sleep assessment methods such as polysomnography (39), some used subjective methods such as Leeds Sleep Evaluation Questionnaire (LSEQ) (40), and some studies incorporated both (41, 42). Post-intervention data only, was reported in some studies (39, 41). Hence, due to the heterogeneity in tools and methods used and strategies taken to report the outcomes, it was impossible to conduct a meta-analysis for this review. Therefore, a narrative approach was taken to describe the accumulated body of evidence.

Results

The PRISMA flow diagram illustrates the process for screening and including eligible articles for this review (**Fig 1**). Removing the duplicates, 2050 articles were screened based on their titles/abstracts, after which 18 papers remained for full-text screening. After excluding articles in which sleep was not assessed as an outcome (n=11), nine more studies were excluded due to supplementation of caffeine in the form of energy drinks (7, 39), including adolescents (43, 44) or recreationally active individuals (45, 46) as participants, reporting no sleep-related outcome (47, 48), and being a study protocol (49). Eventually, 9 articles were evaluated to be eligible for the current systematic review(36-42, 50, 51) .

Characteristics of the included articles are presented in **Table 2**. Included studies were published between 2010 and 2022. All studies included were cross-over trials, had a sample size of 6 to 16 individuals (overall 89 participants). Two studies included both sexes (41, 51), Six studies comprised solely of males (36-39, 42, 50), and one study recruited only female participants (40). Participants' weight ranged from 59.7 ± 3.5 to 87.0 ± 6.0 kg, and their age ranged from 18.78± 1.09 to 28.0± 4.1 years. Participants of five studies were involved in individual sports (38, 39, 41, 42, 51), while those in the remaining four studies were team players (36, 37, 40, 50). Caffeine supplementation was performed between 45 to 75 minutes prior to the engagement of physical activities, including swimming (51), running (36, 38, 40, 42, 50), cycling, basketball (37), or Judo (41). Caffeine was taken at the amount of 3 to 6 mg/kg, once before the prescribed exercise, except for the study of Miller et al., in which the caffeine was ingested once before and once in the middle of the exercise (39). Defining low caffeine intake as less than 100 mg/day, participants in five studies were categorized as low caffeine

consumers (36, 37, 39, 41), 3 as high caffeine consumers (40-42), and one study did not report on the participants' habitual caffeine intake (51). Unfortunately, sleep onset time was not reported in the included trials, but it was reported that caffeine supplement was consumed between 4 to 8 PM, and in one study which assessed the effect of caffeine on daytime nap, it was taken at 2 PM (38).

Sleep Quality:

Four studies examined the effects of caffeine on overall sleep quality (40-42, 51). Three of these studies reported a significant reduction in sleep quality following caffeine ingestion (40-42).

Ali et al. used the Leeds Sleep Evaluation Questionnaire (LSEQ), a 100-mm visual analogue scale, to assess sleep quality. Their findings indicated that caffeine users experienced significantly more periods of restlessness (7.1 ± 2.5) compared to baseline (2.4 ± 1.5) and placebo (3.8 ± 2.3) groups. Additionally, the caffeine group had significantly more periods of awakeness (7.0 ± 2.5) compared to the baseline (2.6 ± 1.6) and placebo (5.4 ± 2.6) groups. Notably, this study included only female participants (40).

Ramos-Campo et al. and Filip-Stachnik et al. employed the Karolinska Sleep Questionnaire (41, 42), which assesses various subscales of sleep quality. Both studies reported significant reductions in sleep quality in the caffeine groups compared to the placebo group, suggesting that the measurement tool may contribute to observed heterogeneity.

In contrast, Vandenbogaerde et al. used a 21-point scale to assess sleep quality and found no clear changes following caffeine consumption (51). However, they noted that their measurement tool had some reliability issues, and other sleep metrics were adversely affected.

Romdhani et al. investigated the effects of caffeine on a 20-minute nap taken after supplementation and found no significant difference in nap quality between the caffeine and placebo groups. In this study, caffeine was consumed at 2 PM, with the nap occurring from 2:10 to 2:30 PM. Sleep quality was rated subjectively using a 100-mm visual analogue scale ranging from 0 (no sleep) to 100 (deep, uninterrupted sleep) (38).

Sleep Duration:

Five studies investigated the impact of caffeine supplementation on sleep duration. Of these, two studies observed a significant reduction in sleep duration (36, 41, 42), while another two reported no significant differences (39, 51). Miller et al. found that the caffeine group slept for 391.4 ± 96.9 minutes, compared to 464.3 ± 48.9 minutes for the placebo group ($p = 0.028$). They also noted that absolute REM sleep duration was shorter in athletes receiving caffeine (39). Vandenbogaerde et al. reported a significant decrease in mean sleep duration, averaging a reduction of 47 minutes (51).

In contrast, Pontifex et al. reported no significant differences in sleep measures (36). Filip-Stachnik et al. and Ramos-Campo et al. also did not find significant results, but it is noteworthy that the caffeine groups had lower mean sleep durations of 402.3 ± 136.3 minutes and 6.6 ± 0.8 hours, respectively, compared to the placebo groups (434.8 ± 119.7 minutes and 7.1 ± 1.4 hours) (41, 42).

Awakening:

Out of six studies assessing the number of awakenings, two reported significant differences (40, 42). Ramos-Campo et al. found that the caffeine group awakened an average of 18.85 times (±7.50) throughout the night, compared to 13.62 times (±7.05) in the placebo group ($p < 0.001$). Additionally, the total time spent awake for the caffeine group was significantly longer at 52.1 ± 23.2 minutes, compared to 29.2 ± 15.4 minutes for the placebo group ($p < 0.005$). However, no significant differences were observed in the average duration of each awakening (42).

These findings are consistent with Ali et al., who reported significantly more periods of awakening in the caffeine group but noted no significant differences regarding the time taken to return to sleep or the difficulty in doing so (40). The remaining four studies found no effects on awakenings when comparing the groups (36, 39, 41, 51).

Sleep Efficiency:

Among the four trials assessing sleep efficiency, two reported a decrease following caffeine supplementation (39, 42). Notably, older athletes (≥ 23 years old) exhibited lower sleep efficiency compared to younger athletes (< 23 years old). Additionally, shorter intervals between caffeine intake and athletic performance 45 minutes before exercise in Ramos-Campo

et al. (42) and 40 minutes into exercise in Miller et al. (39) may contribute to reduced sleep efficiency. The other two studies found no significant differences in sleep efficiency between groups (36, 41).

Sleep Onset Latency and Ease of Falling Asleep:

Sleep onset latency, which measures the time taken to transition from wakefulness to sleep, and ease of falling asleep, reflecting subjective perceptions, are reported together due to their interrelation. Sleep onset latency was measured across several studies (38, 40, 41, 50), while Ali et al. also included ease of falling asleep (39). Two studies reported both measures (40, 41).

Miller et al. found a significant increase in sleep onset latency for the caffeine group (51.1 ± 34.7 minutes) compared to the placebo group (10.2 ± 4.6 minutes) ($p < 0.05$) (39). Vandenbogaerde et al. similarly reported increased time to fall asleep. Although habitual caffeine intake was not specified in Vandenbogaerde et al., it is possible that athletes with lower habitual consumption (<50 mg/day, as noted in Miller et al.) experienced greater delays in sleep onset (51). Miller et al. also did not report weight measurements, which could influence sleep onset latency (39).

Ali et al. found that participants using caffeine took significantly longer to fall asleep (5.9 ± 3.2 minutes) than those in the placebo (2.8 ± 1.5 minutes) and baseline (3.2 ± 2.3 minutes) groups (40). Despite two studies showing no variations in sleep onset latency (39, 41), both Ramos-Campo et al. and Ali et al. reported that the caffeine group experienced significantly greater difficulty falling asleep (40, 42). Ali et al. recorded a difficulty score of 5.9 ± 3.2 for the caffeine group compared to 3.1 ± 1.7 for the placebo and 3.2 ± 2.4 for the baseline group (40). Ramos-Campo et al. reported a lower score for the caffeine group (1.57 ± 0.85) compared to the placebo group (3.43 ± 1.22). It's important to note that one test used higher values to indicate greater difficulty (42), while the other used lower values (41).

Insomnia:

Two studies explored the potential of caffeine supplementation to induce insomnia (37, 50). Raya-González found that caffeine consumption significantly increased insomnia prevalence compared to the placebo group (57% vs. 14%) (37). López-Samanes et al. reported that, while

there was no significant difference between groups, insomnia prevalence in the caffeine group was more than double that of the placebo group (31% vs. 12.5%) (50).

Other Sleep Parameters:

Romdhani et al. assessed the impact of caffeine on nap quality, finding no significant difference between the caffeine and placebo groups. In this study, caffeine was consumed at 2 PM, with the nap occurring from 2:10 to 2:30 PM. Sleep quality was subjectively rated using a 100-mm visual analogue scale ranging from 0 (no sleep) to 100 (deep, uninterrupted sleep) (38).

While one study indicated that caffeine significantly reduced ease of falling asleep, overall time in bed remained unchanged (41, 42). Dreaming was also unaffected by caffeine supplementation (41, 42). Additionally, Ramos-Campo et al. reported a decrease in calm sleep (42). Miller et al. noted increased drowsiness before sleep in the caffeine group (39).

Factors such as alertness, arousal index (39), the time between caffeine/placebo intake and sleep onset, and ease of falling asleep (41)were not significantly impacted by caffeine among athletes.

Discussion

The use of caffeine among athletes to aid in performance and combat fatigue during training or competition has been widely substantiated in the literature (3, 5, 21). Results from this systematic review suggest that caffeine is likely to influence the sleep of athletes following supplementation and have a detrimental impact on sleep quality following its use. However, the overall impact of caffeine and its effect on sleep can manifest in various ways and be different between individuals. Factors such as the timing of caffeine consumption, habitual caffeine use, the dose of caffeine consumed, or a combination of all of these could influence sleep post exercise (20, 24, 25). Further to this gender and other biological markers may also influence the effects of caffeine on sleep (40). This research represents a significant stride in exploring how caffeine affects sleep structure in athletes.

Whilst caffeine is effective in reducing the feeling of tiredness and increasing alertness, both of which may assist sporting performance, the cost might be the disruption of sleep patterns for the athlete post supplementation (13, 52). Understanding the effects of caffeine on sleep parameters in athletes is central to this mechanism; caffeine appears to delay sleep onset, reduce

305 sleep efficiency, and alter sleep architecture, including a reduction in slow-wave sleep, which
306 is critical for physical recovery and cognitive function (53). In addition to its primary effect on
307 adenosine receptors, caffeine affects other neurotransmitter systems, such as the dopaminergic,
308 serotonergic, and noradrenergic pathways, which also contribute to its wake-promoting effects.
309 These interactions not only affect the ability to initiate sleep but may also alter the quality of
310 sleep (54). For athletes, it is important to be aware of caffeine's effects on sleep quality, as sleep
311 is closely linked to performance and recovery (55). The timing of caffeine consumption is
312 critical, as consuming caffeine too close to bedtime can lead to prolonged sleep latency and
313 fragmented sleep, which can compromise the recovery processes essential for optimal
314 performance (41, 42).

315 Individual variability in response to caffeine is another important consideration (56-60).
316 Genetic differences in metabolizing caffeine, particularly variations in the CYP1A2 gene, can
317 lead to significant differences in how long caffeine remains in the body. Slow metabolisers may
318 experience sleep disturbance from caffeine consumed early in the day, whereas fast
319 metabolisers may not experience such effects (56, 57). In addition, habitual use of caffeine can
320 lead to a tolerance, which may reduce the perceived effects on sleep, although this adaptation
321 does not generally negate caffeine's ability to shorten sleep duration and worsen sleep quality
322 (61-64).

323 In the context of athletic performance and recovery, it is important to understand the
324 stimulatory effects of caffeine can be a double-edged sword. While they can enhance alertness
325 and performance, particularly in endurance sports, they can also interfere with the restorative
326 processes of sleep (22, 36, 42). This highlights the need for athletes and their management
327 teams to consider caffeine's ergogenic properties do not interfere with sleep by assessing not
328 only the timing and dosage of caffeine intake but also individual differences in metabolism and
329 tolerance when implementing caffeine into their training and competition routines (22). Hence
330 there should not be a one size fits all approach to caffeine supplementation.

331 The timing of caffeine intake emerges as a critical consideration, with recommendations
332 suggesting that consumption should occur early in the day, ideally more than 6 hours before
333 bedtime, to minimise its sleep-disrupting effects (63, 65). These recommendations are
334 particularly relevant for athletes with slower caffeine metabolism, who may require more
335 careful management of caffeine intake to avoid prolonged sleep latency and disrupted sleep
336 architecture (57). By adjusting caffeine use to individual metabolic profiles and tolerance

levels, and promoting good sleep hygiene practices, athletes may be able to harness the performance-enhancing potential of caffeine while mitigating its risks to sleep quality and recovery processes (57, 63, 65). Where training or performance start times are scheduled later in the day, each athlete's individual tolerance and response to caffeine should be considered prior to supplementation to ensure the timing maximises the performance-enhancing benefits while minimising its adverse effects on sleep (64).

In conclusion, this systematic review has illuminated the complex landscape of caffeine supplementation and its multiple effects on sleep in professional athletes. Our findings highlight that while caffeine is widely used for its ergogenic benefits to sports performance, its effects on sleep are nuanced and influenced by a variety of factors, including the timing of intake, dosage, individual metabolic differences, and habitual consumption patterns. Evidence suggests that, if not carefully managed, caffeine has the potential to disrupt sleep architecture, delay sleep onset, and reduce sleep quality - factors that are critical to an athlete's recovery and longer-term performance (63, 65).

Future research should aim to expand the existing evidence base with larger, more diverse study populations and longitudinal designs to explore the long-term effects of caffeine on sleep and recovery in athletes. In addition, further investigation into the interplay between genetic factors, habitual caffeine consumption and the timing of caffeine intake may provide deeper insights into personalised dietary and supplementation strategies for athletes.

Strengths and limitations

This systematic review critically evaluates the impact of caffeine supplementation on sleep parameters in professional athletes. The methodology includes a comprehensive search of multiple databases and adherence to strict inclusion criteria for randomized controlled trials, which minimizes publication bias and ensures the relevance and quality of the evidence reviewed. The inclusion of both objective and subjective measures of sleep allows for a detailed examination of the effects of caffeine on sleep. The conclusions of the review are limited due to the heterogeneity of the included study designs and the small number of studies that met the strict inclusion criteria. Having limited literature to systematically review makes it problematic to draw clear conclusions and to perform a meta-analysis. In addition, the possibility of publication bias and differences in sleep assessment methods may also influence the results. In addition, most studies have focused on short-term effects and have included a limited demographic group, mainly young male athletes, which further limits the generalisability of

the findings. Future research should aim to address the limitations by standardising sleep assessment methods, exploring the long-term effects of caffeine, and including a broader demographic to fully understand the implications of caffeine supplementation on sleep in athletes. Also, a key limitation of the included trials is the inconsistent reporting of the timing of caffeine ingestion in relation to exercise and sleep. Most studies lacked precise details, which limits understanding of caffeine's full impact on athletes' sleep. Future research should address this gap by providing clearer data on the timing of caffeine administration and its effects on sleep.

Conclusion

Caffeine is highly consumed by athletes during training and competitions, and it may affect sleep through various mechanisms. These mechanisms may be manipulated by specific factors such as caffeine dose, timing of caffeine intake, habitual caffeine consumption, etc. Further research with unified sleep measurement tools are required to be able to perform a meta-analysis and provide guidance around the optimum timing and dose of caffeine in relation to effects on sleep architecture.

Author Contributions:

ZSH and MB contributed to the conception of this study and performed the systematic search through the databases. CL and MB screened and selected the eligible studies, performed the data extraction. They also assessed the quality of the included studies. ZSH interpreted the results. ZSH and JB drafted the manuscript. JB supervised this research project and revised the manuscript critically for important intellectual content. All authors approved of the final draft of the manuscript and agreed to be accountable for all aspects of the work.

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Conflict of Interest:

None to be declared.

Data Availability Statement:

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398 All relevant data is available within the manuscript and its supplementary files.

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Figure 1: PRISMA flowchart.

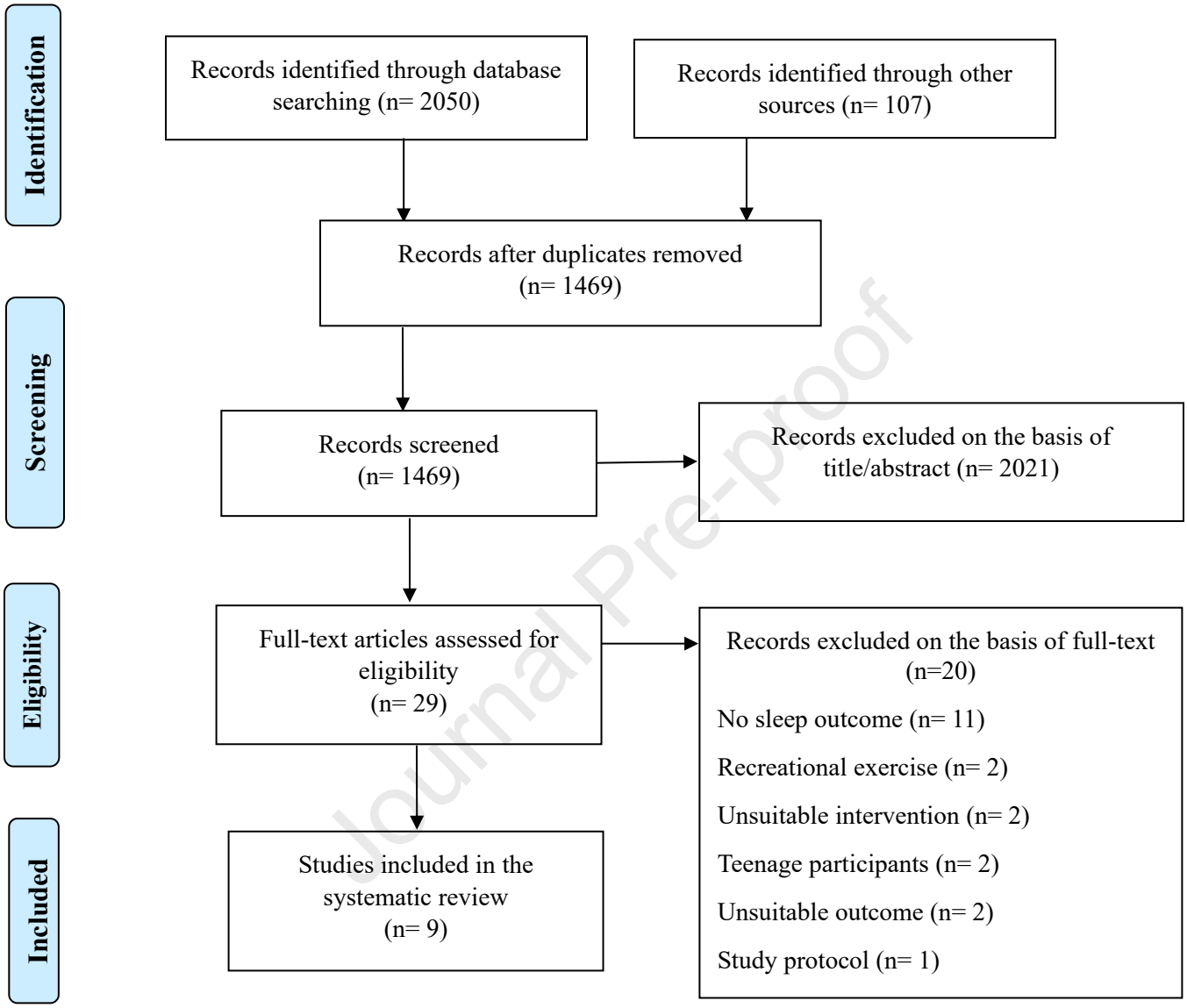

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Figure 2: Risk of Bias assessed by Revised Cochrane risk-of-bias tool for randomized crossover trials (RoB-2).

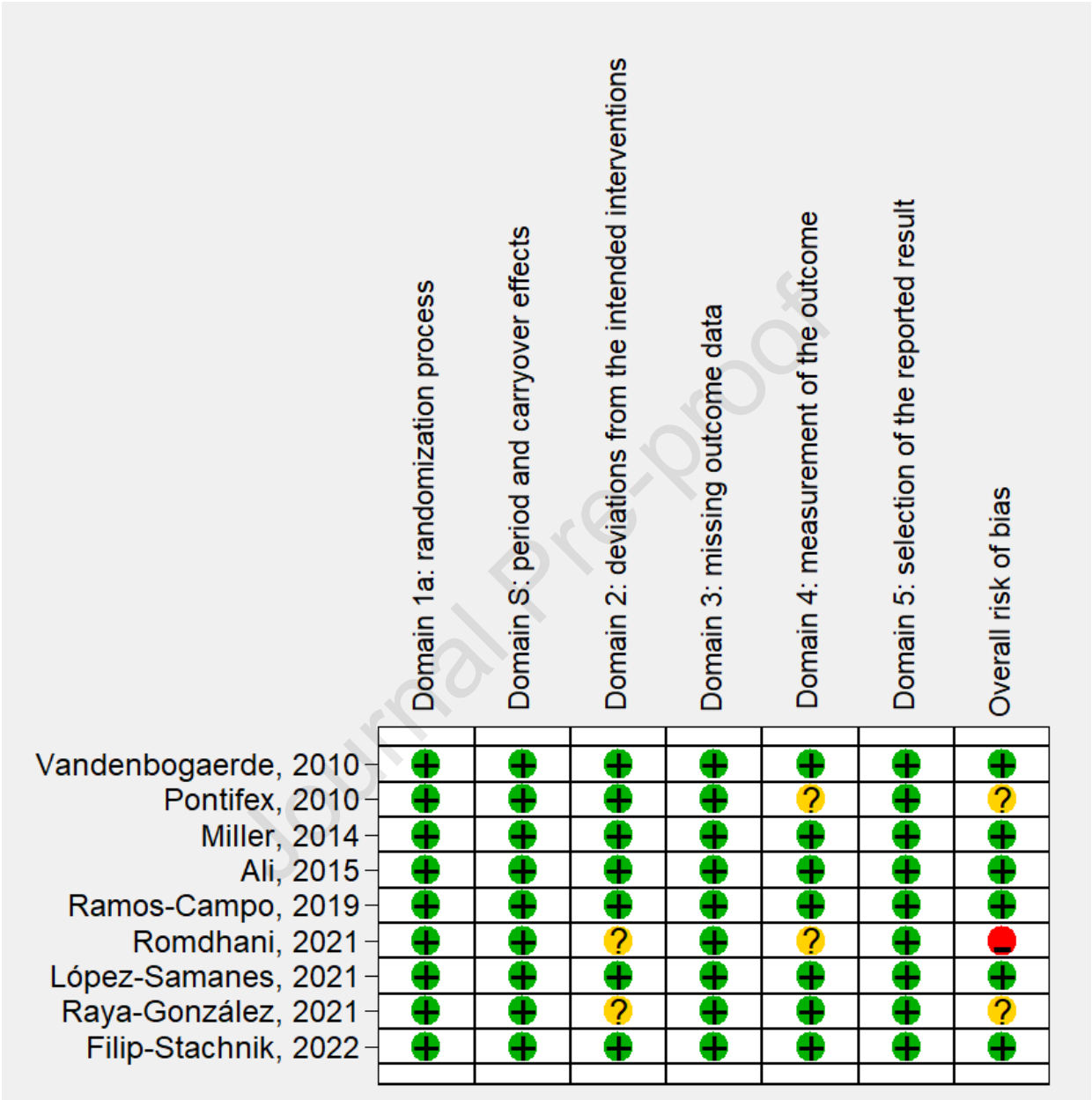

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Table 1: PICOS criteria for inclusion and exclusion of studies.

Parameter	Inclusion criteria	Exclusion criteria
***P*articipants**	Adult athletes aged >18 years old	Non-athlete adults, children and adolescents; older adults, individuals engaged in recreational activities and exercise
***I*ntervention**	Caffeine tablet, capsule or powder, consumed prior to physical activity or athletic performance	Caffeine supplementation in the form of energy drinks, in combination with other nutrients and amino acids affecting sleep
***C*ontrol/comparator groups**	Placebo	-
***O*utcomes**	*Objectively assessed*: polysomnography, actigraphy *Subjectively assessed*: sleep duration, sleep quality, sleep efficiency, time in bed, sleep time (sleep onset), wake time (sleep offset), number of awakening, average time of each awakening, sleep latency, insomnia	Sleep parameters not being reported
***S*tudy design**	All open-label or single, double, and triple-blind RCTs	Case reports; observational studies; animal studies; review, systematic review, or meta-analysis articles; protocols, editorials; RCTs without available full-text

RCTs: randomized controlled trials.

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Table 2. Main features of studies included in the systematic review.

Row	Author (year)	Country	N	Mean weight (kg)	Sex (%male)	Mean age (years)	Caffeine dose (mg/kg)	Athletic population	Prescribed exercise	Time from caffeine intake to exercise	Proximate time of caffeine intake	Mean habitual caffeine intake	Placebo	Main findings
1	Vandenbogaerde, 2010	New Zealand	9	73 ± 4.5	both sexes (67%)	23.5 ± 2	5	Elite sprint and middle-distance swimmers	stretching followed by 1–2 km of swimming	75 min	4 to 7 PM	NA	custard powder	Non-significant changes in number of awakenings and sleep quality; Sleep duration ↓ Time to fall asleep ↑
2	Pontifex, 2010	Australia	10	75.5 ± 4.52	male	21.6 ± 2.6	6	Football, soccer and field hockey players	4 min jog followed by 4*20 m sprint, reactive agility time trials 10*10.2 m, 7 min of active recovery, repeated sprint test 6*20 m, 5 min of active recovery, and another reactive agility time trial 10*10.2 m	60 min	4 to 7 PM	58±43 mg/day	artificial sweetener	Non-significant change in number of awakenings, sleep efficiency, wake time, and sleep duration
3	Miller, 2014*	Australia	6	NA	male	27.5 ± 6.9	3 * 2 times	cyclists/ triathletes	an afternoon training session (cycling 80 min; 65 % VO2max) followed by a 5 kJ/kg cycling time trial	60 min prior and 40 min into the training session	twice at 4 PM and 5:40 PM	<50 mg/day	lactose	Non-significant changes in number of awakenings, arousal index, sleep onset latency, and duration of different sleep stages; sleep efficiency ↓ total sleep time ↓ absolute REM sleep duration ↓ wake after sleep onset ↑
4	Ali, 2015	New Zealand	10	59.7 ± 3.5	female	24 ± 4	6	athletes in team sports (soccer, hockey and netball)	intermittent exercise: 6*15 min blocks of treadmill running with each block consisting of 2-min periods of varying speeds, followed by 1 min at 60% V˙O2 max, and a 2-min walk (4 km/h).	45 min	around 5 PM	150 mg/day	artificial sweetener	Non-significant changes in periods of awakeness; difficulty, alertness and duration of awakenings; alertness; balance/ coordination on awakenings; Getting to sleep harder ↑ Getting to sleep slower ↑ drowsiness level before sleep ↑ quality of sleep ↓
5	Ramos-Campo, 2019	Spain	15	64.6 ± 9.8	male	23.7 ± 8.2	6	middle-distance runners	15min warm up (running at low intensity, joint mobility, dynamic stretching, and progressive running sets), a	45 min	around 7 PM	411 mg/day	sucrose	Non-significant changes in sleep latency, total time in bed, sleep time, average time of each awakening, amount of

									countermovement jump test, and 2 min later, 800-m running time-trial test					dreaming, ease of waking up, and sleep throughout the time allotted; sleep efficiency ↓ sleep quality ↓ calm sleep ↓ feeling refreshed after awakening ↓ ease of falling asleep ↓ actual wake time ↑ number of awakenings ↑
6	Romdhani, 2021	Tunisia	9	68.44 ± 6.19	male	18.78 ±1.09	5	judokas	The Running-Based Anaerobic Sprint Test including 6 maximal all-out repeated sprints of ~6 to 7 seconds interspersed with 10-second recovery periods	50 min	2 PM	≤80	cellulose & starch–based	Non-significant change in subjective sleep quality (nap) scores
7	López-Samanes, 2021	Portugal	16	71.2 ± 9.7	male	28.0 ± 4.1	3	futsal players	countermovement jump, 20-m sprint test, and a futsal kicking velocity and accuracy test	60 min	between 5 to 7 PM	< 1 mg/kg/day	cellulose	Insomnia ↑
8	Raya-González, 2021	Spain	14	87.0 ± 6.0	male	21.0 ± 2.0	6	basketball players	20-m sprints, countermovement jumps, Lane Agility Drill trials, and a repeated-sprint-ability test	60 min	between 6:30 to 8 PM	<100 mg/day	sucrose	Insomnia ↑
9	Filip-Stachnik, 2022	Poland	9	71.75 ± 13.3	both sexes (55.5%)	22 ± 3	3	judokas	warm-up followed by randori training session (3 fighting bouts of 4 min, with 10 min of rest between fighting bouts)	60 min	around 6 PM	2.8±2.4 mg/kg/day	cellulose	Non-significant changes in sleep duration, time in bed, sleep efficiency, wake after sleep onset, number of awakenings, time between substance intake and sleep onset, calm sleep, ease of falling asleep, amount of dreaming, ease of waking up, feeling refreshed after awakening, sleep throughout the time allotted; sleep quality ↓

N: Number of Participants, NA: Not available; m: meter, min: minute; % V˙O2 max: *Maximum rate of oxygen used during the exercise.*

Proximate times of caffeine intake were not directly reported, and was approximately estimated based on the information provided in the publications.

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**In the study by Miller et al., caffeine was administered in the dose of 3 mg/kg, once before and once in the middle of the exercise.*

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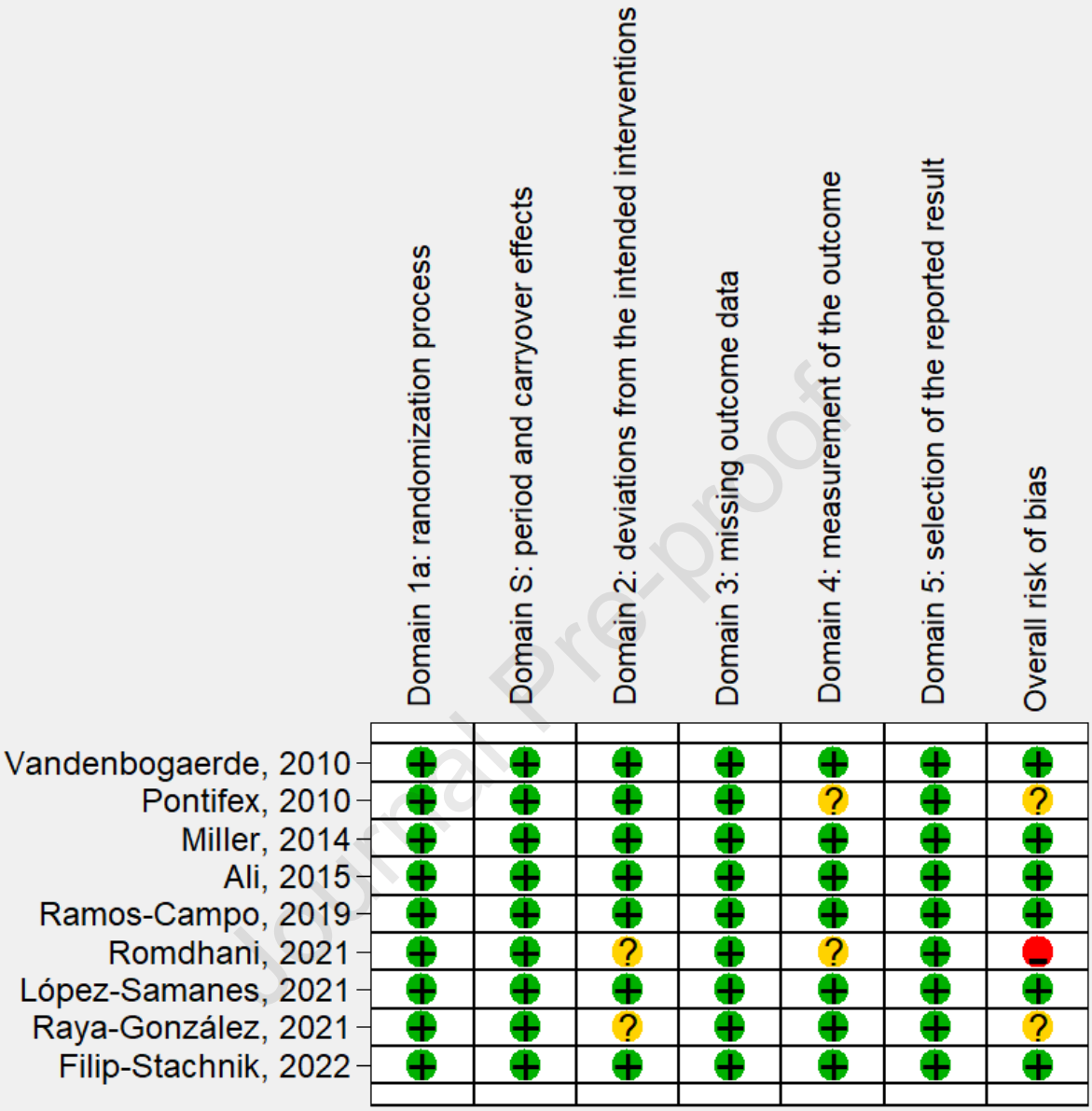

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